Test Data Threshold = 0.5

Actual label	Predicted score	Predicted label
1	0.85	1 ✓
0	0.6	0 ✓
1	0.7	1 ✓
1	0.4	0 ✓
0	0.55	0 ✓
1	0.5	1 ✓
0	0.65	1 ✓
0	0.35	0 ✓
1	0.6	0 ✓
0	0.2	0 ✓

Confusion Matrix
↓
Accuracy
Precision
Recall Sensitivity
F1 - score
Compute ROC curve
threshold of
0.7, 0.5, 0.4, 0.2

=> >= 0.5 -

	Original Positive	Original Negative
Predicted Positive	TP = 4	FP = 0
Predicted Negative	FN = 3	TN = 3

$$\text{Accuracy} = \frac{(TP+TN)}{\text{total}} = \frac{4+3}{10} = 0.7$$

$$\text{Precision} = \frac{TP}{(TP+FP)} = \frac{4}{4+0} = 1$$

$$\text{Sensitivity (Recall)} = \frac{TP}{(TP+FN)} = \frac{4}{(4+3)} = \frac{4}{7} = 0.571$$

Page No.:
Date: / /

$$\text{Specificity} = \frac{TN}{(TN+FP)} = \frac{3}{(3+0)} = 1$$

$$\underline{F1} = \frac{2TP}{2TP+FP+FN} = \frac{2\times 4}{2\times 4+0+3} = \frac{8}{11}$$

$$F1 = \boxed{0.727} = \frac{1}{\frac{1}{1}+\frac{1}{0.57}} \quad \frac{1}{\frac{2}{2.75}} = \frac{1}{1.37}$$

$$= \underline{0.726}$$

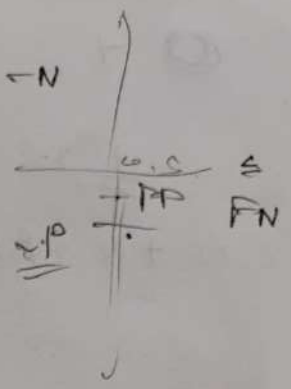
Actual Label	Predicted Score	Predict Label	
1	0.85	1	TP ✓ = TP ✓
0	0.6	1	FP ✓ = FP ✓
1	0.7	1	TP ✓ = TP ✓
1	0.4	0	FN ✓ = FN ✓
0	0.65	1	FP ✓ = FP ✓
1	0.5	0	FN ✓ = FN ✓
0	0.65	1	FP ✓ = FP ✓
0	0.35	0	FN ✓ = FN ✓
1	0.6	1	TP ✓ = TP ✓
0	0.2	0	FN ✓ = FN ✓

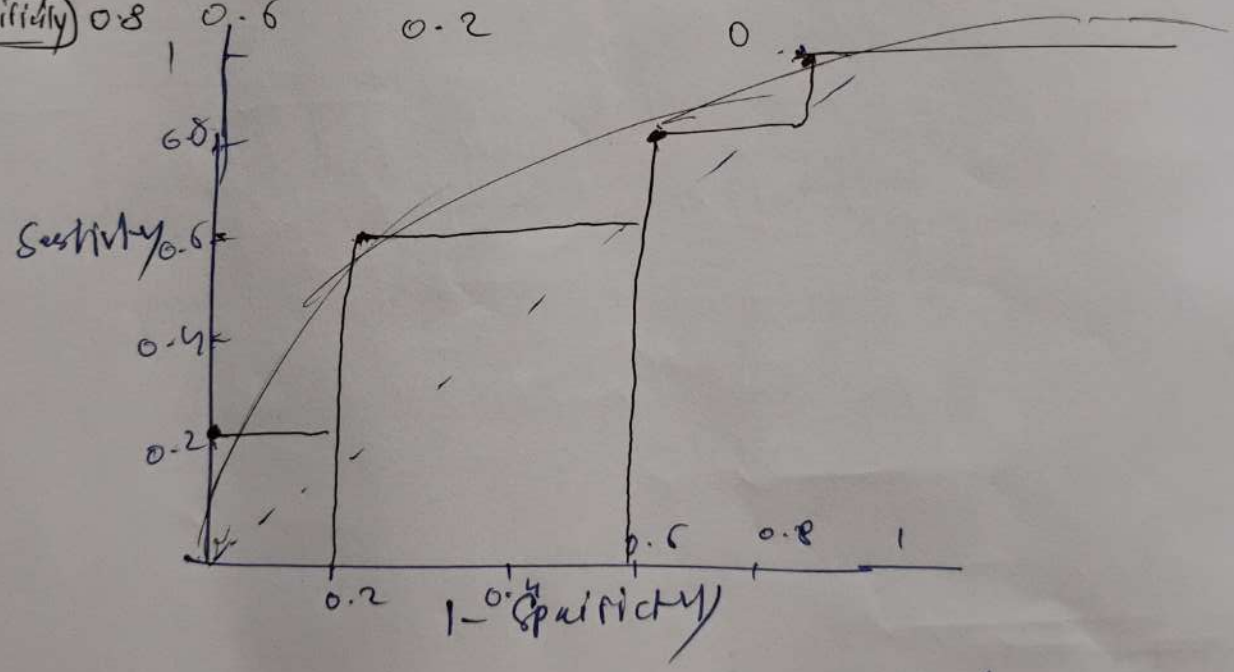
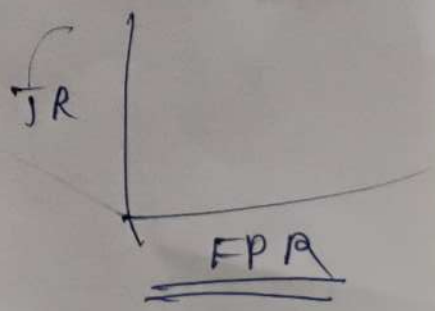
~~TP = 3 TN = 0~~
~~FN = 4 FP = 3~~

TP = 3 TN = 2
FN = 2 FP 3

Teacher's Signature: . . . . .

0 ≤ 0.2	0.4	0.5	0.7
TP	TP	TP	TP
FP	FP	TN	TN
TP	TP	TP	FN
TP	FN	FN	FN
FP	FP	TN	TN
TP	TP	FN	FN
FP	FP	FP	TN
FP	~~TN~~	TN	TN
TP	TP	TP	FN
TN	TN	TN	TN
TP = 5	TP = 4	TP = 3	TP = 1
FP = 4	FP = 3	FP = 1	FP = 0
TN = 0	TN = 2	TN = 4	TN = 5
FN = 0	FN = 1	FN 2	FN = 4
Sensitivity 1	0.8	0.6	0.2
~~Sp~~ Specificity 0.2	0.4	0.8	0
1 - Specificity 0.8	0.6	0.2	0





Sensitivity 1
1 - Specificity